

SMART CLASSROOM BELL SYSTEM USING LPC2148

1. TITLE PAGE

Project Title

Smart Classroom Bell System Using LPC2148

Course Name

Arm lab

Team Number

Team – __2__

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2. ABSTRACT

The Smart Classroom Bell System is an automatic bell indication system developed using the LPC2148 ARM7 microcontroller. The system uses the Real Time Clock (RTC) to maintain accurate timing and automatically turns ON an LED at scheduled times.

A 16×2 LCD is used to display the current time and system status. UART communication is provided to update bell timings through serial communication. An external interrupt switch is also included for manual bell operation during emergency situations.

This project reduces manual work and improves timing accuracy in schools and colleges.

3. INTRODUCTION

In schools and colleges, bells are usually operated manually. Manual operation can lead to mistakes and incorrect timing. Sometimes the bell may ring late or early due to human error.

To solve this problem, an automatic classroom bell system is developed using embedded systems technology. The LPC2148 microcontroller controls the entire system and automatically indicates the bell timing using an LED.

The RTC module keeps track of the current time. The LCD displays the system information. UART communication allows the user to update timings easily. An external interrupt switch is used for manual operation when required.

This project helps in reducing human effort and improving accuracy.

4. PROBLEM STATEMENT

In educational institutions, classroom bells are commonly operated manually. Manual operation requires continuous monitoring and may cause timing errors.

There is a need for an automatic system that can:

- Maintain accurate timing
- Automatically indicate bell timing
- Allow easy schedule updates
- Support manual operation during emergencies

The Smart Classroom Bell System solves this problem using the LPC2148 ARM7 microcontroller.

5. SYSTEM OVERVIEW

Inputs

- RTC Module
- UART Communication
- External Interrupt Switch

Processing Unit

- LPC2148 ARM7 Microcontroller

Outputs

- LED Indicator
- LCD Display
- UART Serial Monitor

Working Principle

The RTC module continuously maintains the current time. The LPC2148 microcontroller checks the RTC timing and compares it with the scheduled bell timing.

When the current time matches the scheduled time:

- The LED turns ON
- The LCD displays “BELL RINGING”

UART communication is used to update the bell schedule. The external interrupt switch is used for manual bell indication.

6. PROBLEM ANALYSIS

Features

1. Automatic bell indication
2. Accurate RTC timing
3. LCD status display
4. UART schedule updating
5. Manual override operation
6. Low-cost implementation

Assumptions

- Proper power supply is available
- RTC maintains correct time
- LCD and UART work properly

Peripherals Used

- RTC
 - UART0
 - GPIO
 - I2C LCD
 - External Interrupt
-

7. HARDWARE DESCRIPTION

LPC2148 MICROCONTROLLER

The LPC2148 is a 32-bit ARM7 microcontroller developed by NXP Semiconductors.

Features

- 512 KB Flash Memory
- 40 KB RAM
- RTC Support
- UART Communication
- I2C Interface
- GPIO Pins

Role in Project

- Controls the entire system
 - Reads RTC timing
 - Controls LED indication
 - Handles UART communication
 - Drives LCD display
-

RTC MODULE

The RTC module maintains the current time.

Role

- Provides accurate timing
 - Controls automatic scheduling
-

LED INDICATOR

An LED is used instead of a buzzer.

Role

- Indicates bell timing
 - Glows during scheduled periods
-

LCD DISPLAY

A 16×2 LCD displays:

- Current time
 - Bell status
 - System messages
-

UART COMMUNICATION

UART communication is used to:

- Update bell schedules
 - Send system messages
-

EXTERNAL INTERRUPT SWITCH

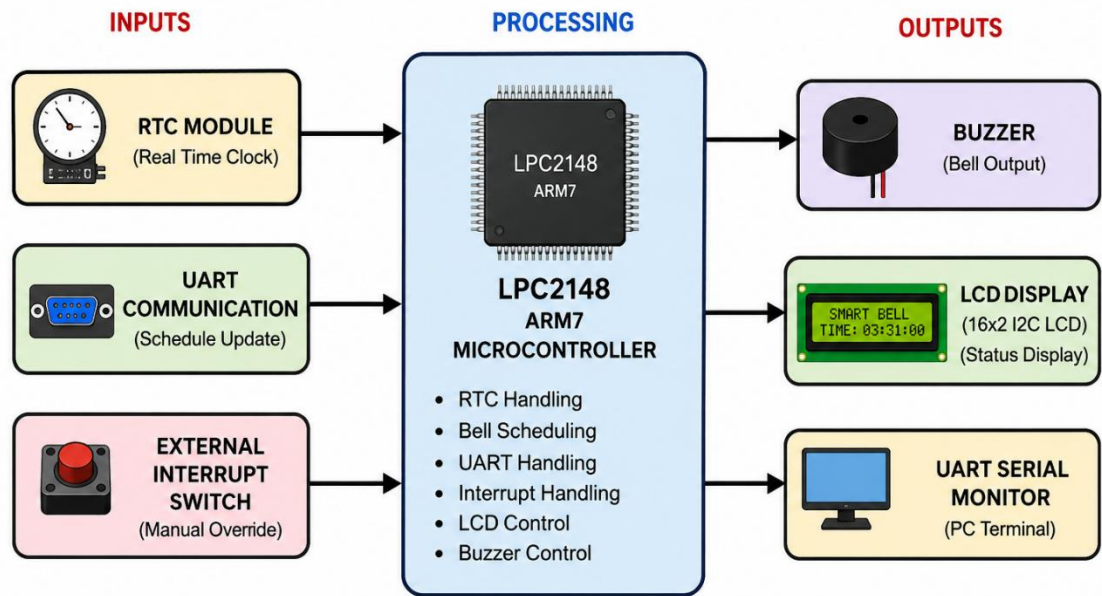
The interrupt switch is connected to EINT0.

Role

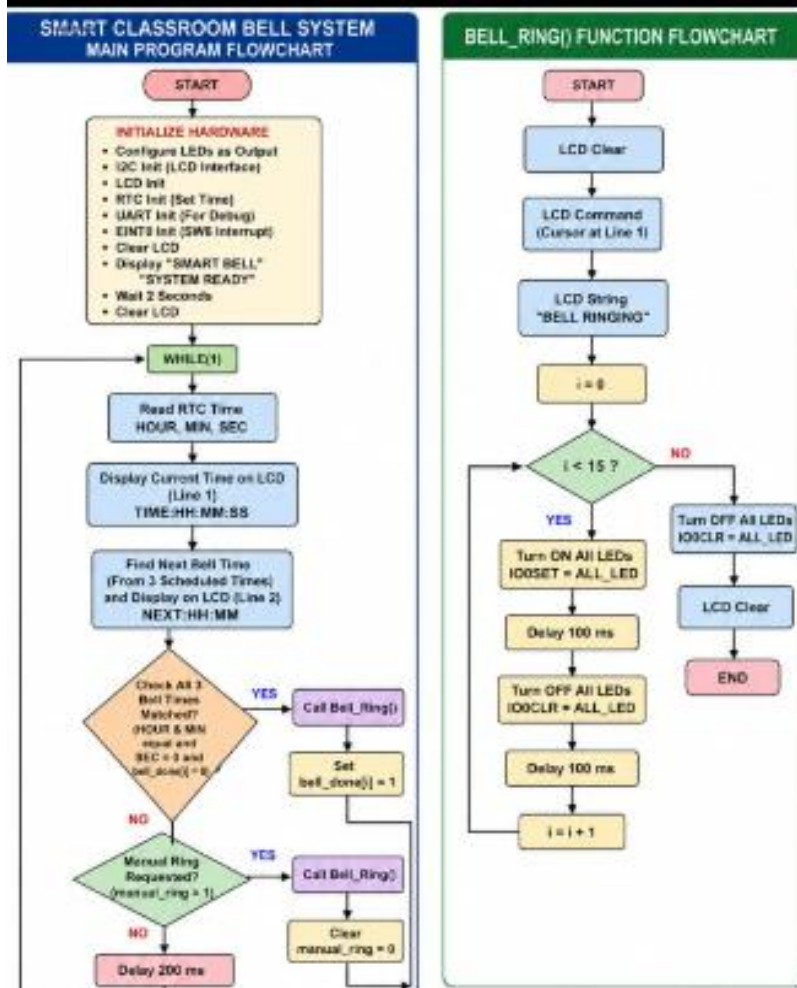
- Manual bell operation
 - Emergency override
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7. BLOCK DIAGRAM

SMART CLASSROOM BELL SYSTEM BLOCK DIAGRAM



FLOWCHART



9. ALGORITHM / FLOWCHART

Algorithm

1. Start the system
2. Initialize RTC, UART, LCD, and GPIO
3. Read current RTC time
4. Display time on LCD
5. Compare RTC time with scheduled time
6. If time matches:
 - Turn ON LED
 - Display "BELL RINGING"

7. Check UART for schedule updates
8. Check interrupt switch
9. Repeat continuously

10. INTERFACING DETAILS

Component	LPC2148 Pin
LED	P0.16
UART TXD0	P0.0
UART RXD0	P0.1
I2C SDA	P0.2
I2C SCL	P0.3
Interrupt Switch	P0.14

11. CALCULATIONS & SPECIFICATIONS

RTC Configuration

RTC values used:

- PREINT = 0x01C8
- PREFRAC = 0x61C0

These values generate a 1-second clock.

UART Baud Rate

- Baud Rate = 9600 bps

LED Specifications

- Operating Voltage = 3.3V
 - Current Limiting Resistor = 220Ω
-

12. IMPLEMENTATION

Hardware Used

- LPC2148 Development Board
- 16×2 LCD
- LED
- Push Button Switch
- UART Cable

Software Used

- Keil uVision IDE
- Embedded C
- Flash Magic
- Proteus

Steps

1. Write Embedded C program
 2. Compile code using Keil
 3. Generate HEX file
 4. Upload using Flash Magic
 5. Test RTC operation
 6. Verify LED indication
 7. Check UART communication
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13. RESULTS

Observed Results

1. RTC displayed correct time
2. LED glowed at scheduled time
3. LCD displayed system status correctly
4. UART updated schedules successfully
5. Interrupt switch worked properly

The project successfully automated classroom bell indication using LPC2148.

14. CONCLUSION

The Smart Classroom Bell System using LPC2148 was successfully implemented. The system automatically indicated bell timing using RTC scheduling and LED indication.

The project demonstrated RTC interfacing, UART communication, LCD interfacing, and interrupt handling using embedded systems.

The system is simple, reliable, and useful for classroom automation.

15. FUTURE SCOPE

1. Wi-Fi based control
 2. Mobile application support
 3. Voice announcement system
 4. Multiple schedule storage
 5. Battery backup system
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16. REFERENCES

Books

1. Embedded Systems – Raj Kamal
2. ARM System Developer's Guide

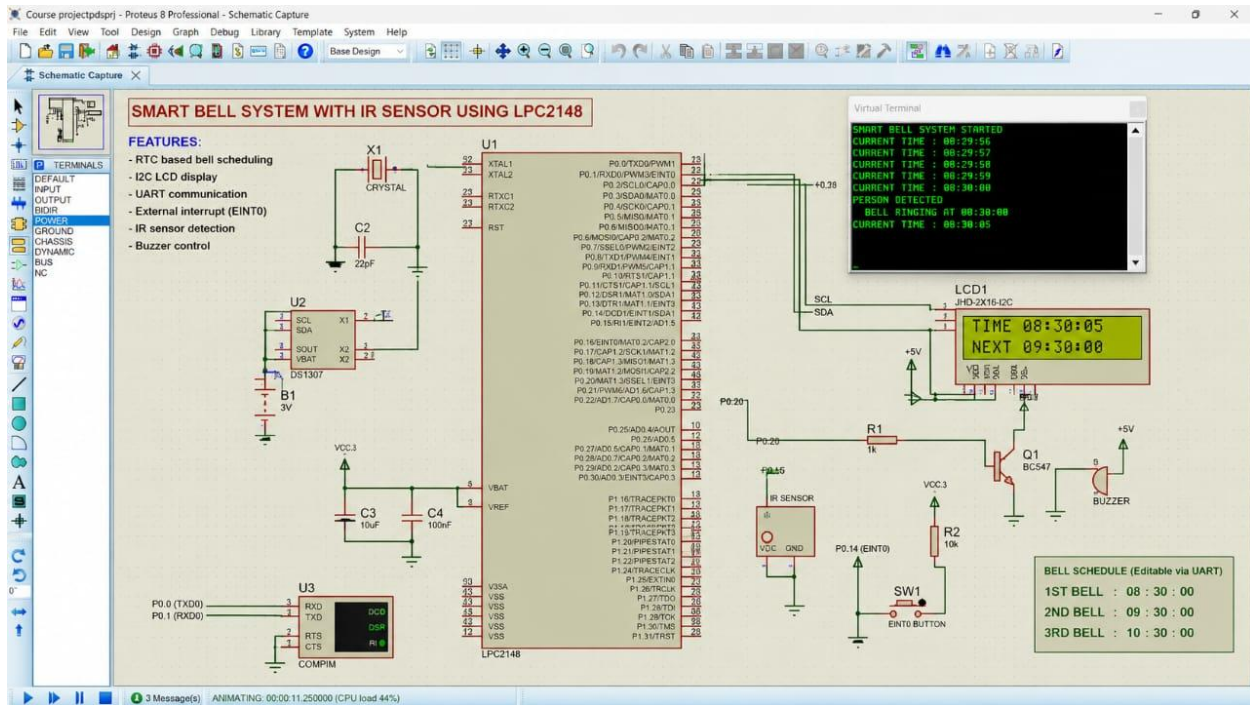
Datasheets

1. LPC2148 Datasheet
2. LCD Datasheet

Websites

1. NXP Semiconductors
2. ARM Developer Documentation

17 .PROTEUS SIMULATION



Output :

